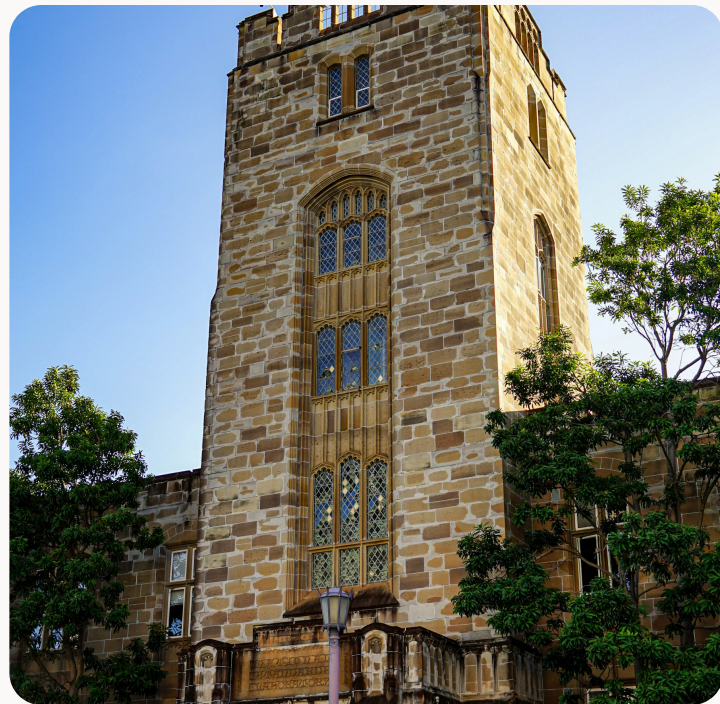


Does spaced retrieval improve first-year statistics learning?

An illustrative field study of low-stakes retrieval and durable learning

Illustrative Learning Sciences Group August 2026





Research question

Does adding brief, spaced retrieval prompts to weekly tutorials improve delayed conceptual performance relative to guided review?

BACKGROUND

Immediate fluency can conceal fragile understanding

Students may perform well during guided practice yet struggle to retrieve the same concepts after a delay.

The instructional challenge is not exposure alone; it is making knowledge accessible when support is removed.

01	Worked examples	High support
02	Independent retrieval	Effortful access
03	Delayed application	Durable use

Prior evidence supports practice testing, but implementation conditions matter

Broad utility

Practice testing has been rated a high-utility learning technique across varied contexts.

Positive synthesis

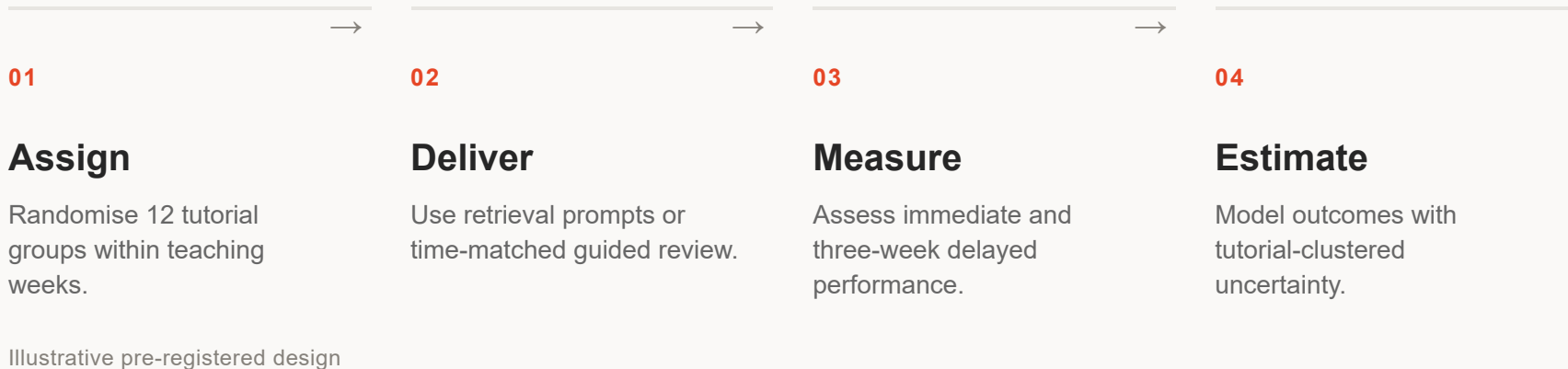
A meta-analysis found practice tests outperform restudy and other non-testing conditions.

Open implementation question

Course context, feedback, timing, and dosage may shape the realised effect.

Dunlosky et al. (2013), doi:10.1177/1529100612453266 · Adesope et al. (2017), doi:10.3102/0034654316689306

A tutorial-level field experiment balances credibility and feasibility



Retrieval should strengthen access to later application

Spaced, effortful retrieval with feedback

Accessible knowledge

Concepts are easier to retrieve after a delay

Calibrated confidence

Feedback exposes gaps before summative assessment

Transfer opportunity

Retrieved concepts can be applied to unfamiliar problems

Illustrative mechanism; the proposed links do not establish causality.

The illustrative sample is balanced at baseline

Measure	Retrieval	Guided review	Std. difference
Students	148	151	—
Prior quantitative score	72.1	71.6	0.03
Weekly attendance	83%	82%	0.02
Baseline concept score	54.8	55.2	−0.02

Unit of assignment · Tutorial group; illustrative analysis accounts for clustering.

RESULT 1 · PRIMARY OUTCOME

The retrieval group performs eight points higher after three weeks

Guided review



64%

Retrieval



72%

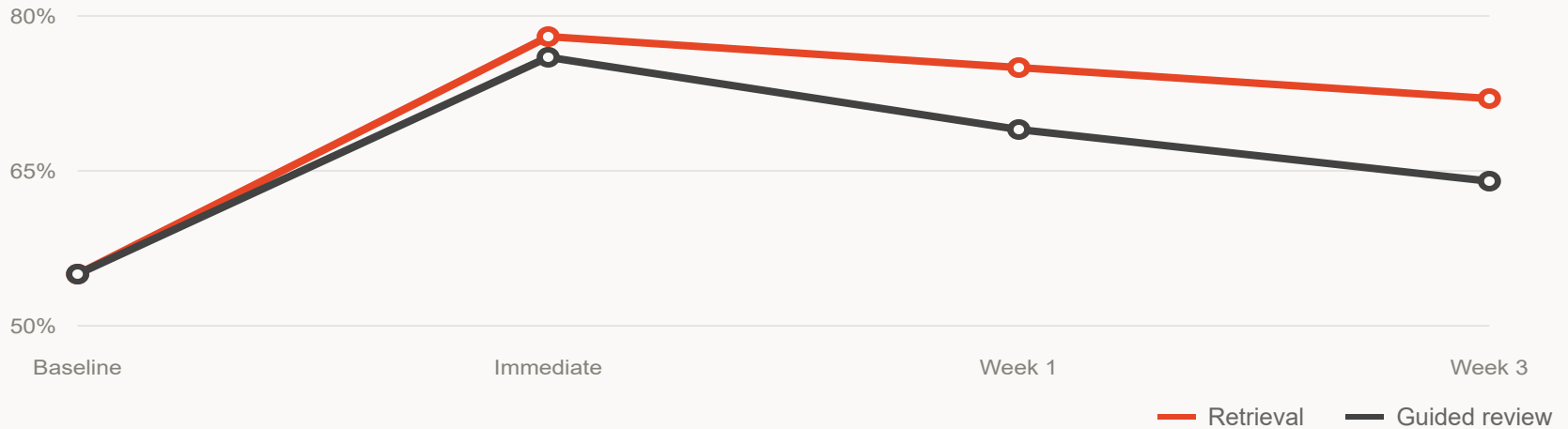
ESTIMATED DIFFERENCE

+8.1 percentage points; illustrative 95% CI [3.2, 13.0].

Illustrative delayed conceptual assessment

RESULT 2 · DURABILITY

The initial gap widens as support is removed



Illustrative mean conceptual score by assessment point

The adjusted effect remains statistically meaningful

Outcome	Estimate	Clustered SE	95% CI	p
Delayed concept score	+8.1 pp	2.3	[3.2, 13.0]	.004
Immediate score	+1.9 pp	1.8	[-1.9, 5.7]	.303
Calibration error	-6.4 pp	2.5	[-11.7, -1.1]	.022

Illustrative linear models with baseline covariates and tutorial-clustered standard errors.

The pattern is consistent with a durability mechanism

WHAT THE RESULTS SUPPORT

A delayed advantage

The groups are similar immediately after instruction, but diverge when students must retrieve concepts after a delay.

WHAT REMAINS UNCERTAIN

The active ingredient

The design does not isolate spacing, retrieval effort, feedback, or their interaction.

LIMITATIONS

Four limitations bound the strength and reach of the conclusion

01	Single course context	External validity
02	Tutorial-level assignment	Limited clusters
03	Short follow-up	Three weeks
04	Composite intervention	Mechanism unresolved

IMPLICATIONS

The practical next step is a measured extension, not immediate scale-up

For teaching

Embed brief retrieval with feedback in high-enrolment tutorials.

For evaluation

Retain delayed outcomes and report implementation fidelity.

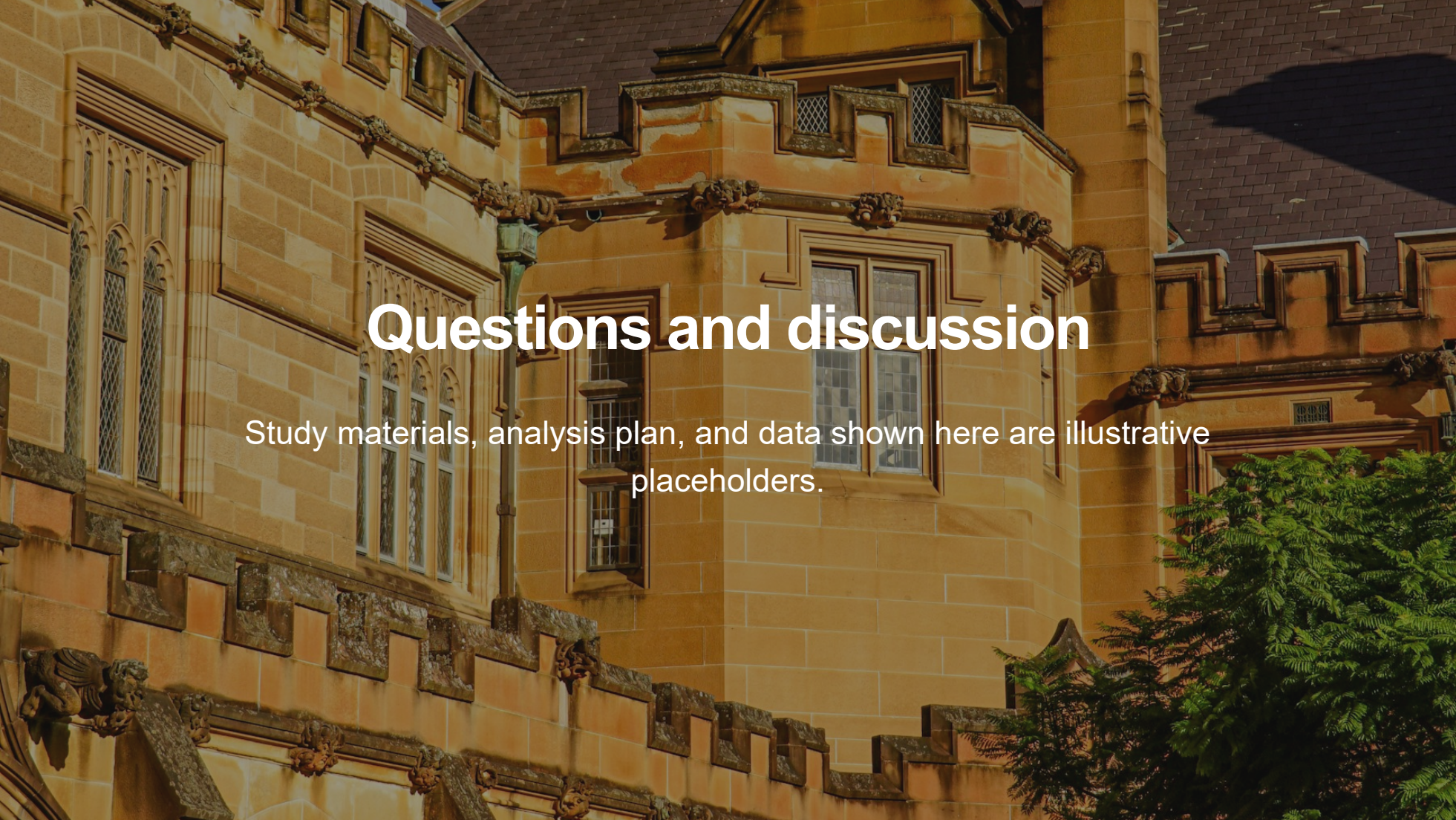
For research

Separate spacing, feedback, and retrieval effort in the next design.

Conclusion

In this illustrative study, spaced retrieval improves delayed conceptual performance while leaving immediate scores broadly similar.

The finding supports a larger, multi-course replication focused on durability and mechanism.



Questions and discussion

Study materials, analysis plan, and data shown here are illustrative placeholders.